WASP-12b

R-0357

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_231126` · created 2026-07-18T11:39:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460274.9454 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.044
Transit depth (Rp/Rs)^2	0.00014 +/- 0.01 [%]
Orbital Inclination (inc)	84.42 +/- 2.93
Airmass coefficient 1 (a1)	3.31 +/- 2.33
Airmass coefficient 2 (a2)	-1.07 +/- 0.76
Scatter in the residuals of the lightcurve fit is	70.51 %
Best Comparison Star	None
Optimal Aperture	1.46
Optimal Annulus	3.0
Transit Duration (day)	0.1146 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.044 vs 0.1178 (deviation 2.65σ)
Duration fit vs published	0.1146 d vs 0.125 d (deviation 1.6σ)
Mid-time O–C	-0.5 min
Depth SNR	0.0
Residual scatter	70.51 %

Light curve

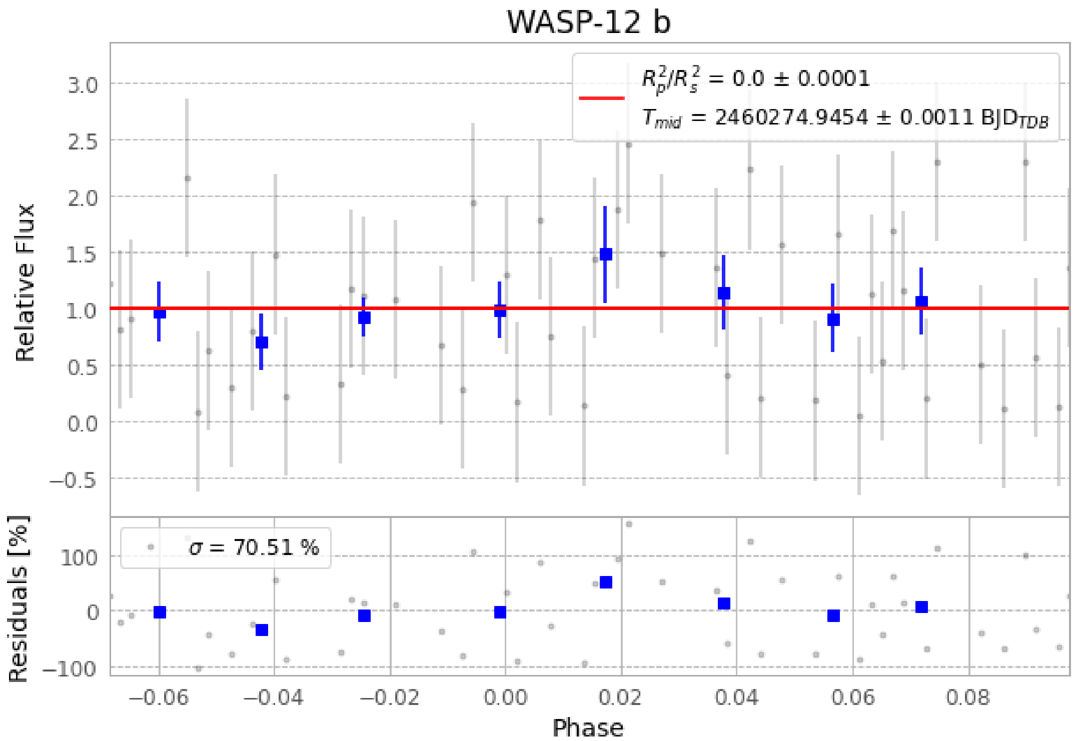

Figure 1 — FinalLightCurve_WASP-12 b_2023-11-26.png (SHA-256 ad471576679f1827…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [338, 314], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 34e996bc927ed8ee…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da`

Data provenance

88 FITS frames (58 MB), `WASP-12231126084515.FITS` → `WASP-12231126130615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231126.log (9528a206c32b…), FinalLightCurve_WASP-12 b_2023-11-26.png (ad471576679f…), FinalLightCurve_WASP-12 b_2023-11-26.csv (3e6e7deaaa53…)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)